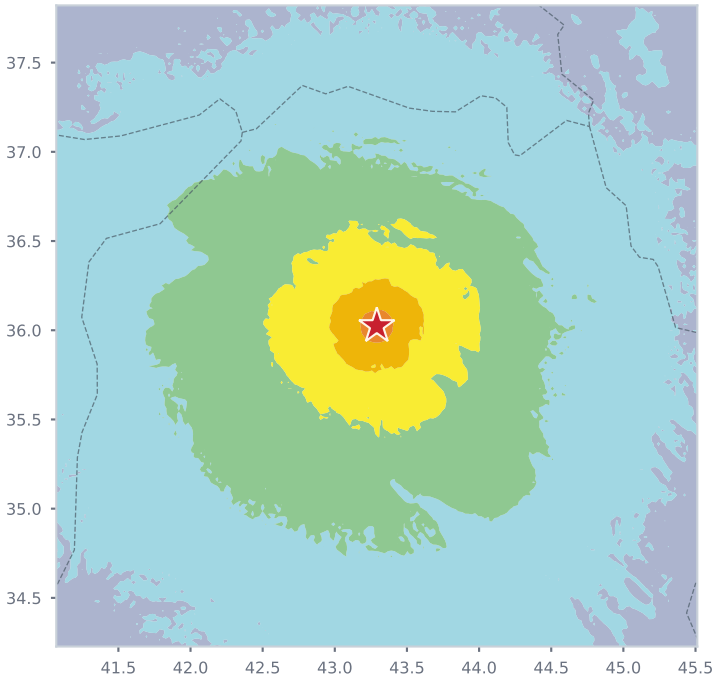
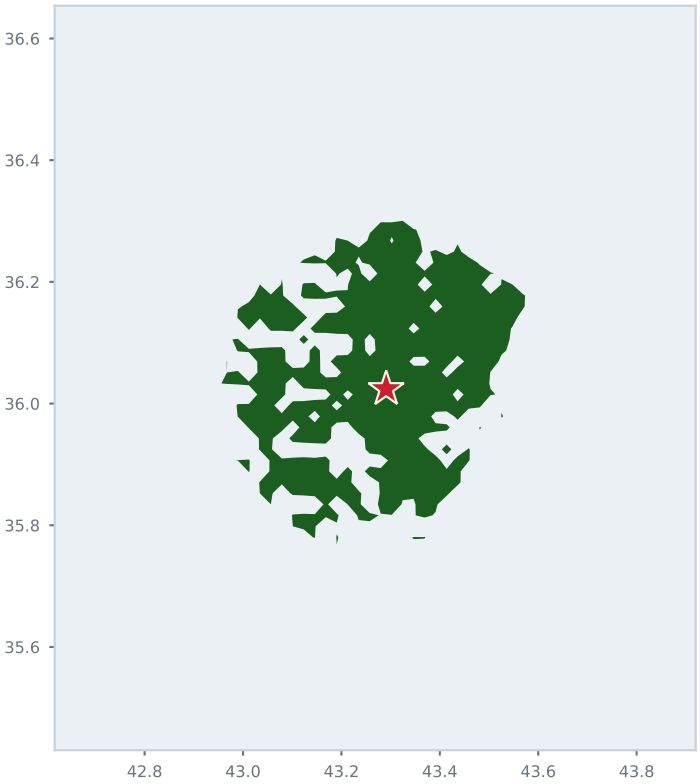
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity



Expected damage grade (IMS-25)



Damage alert

RED Extensive destruction across the area

Provisional **Day occupancy**

Buildings heavily damaged (DG3 or worse)

5.7 thousand to 61 thousand

Central estimate 18 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

3.6 million

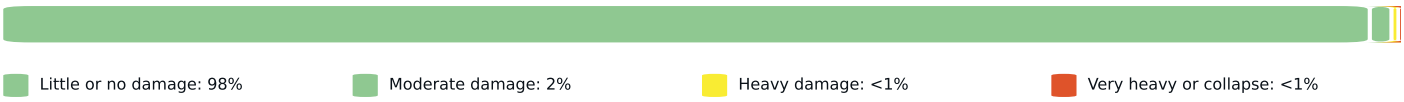
21% of them inferred from built-up area, not observed footprints.

People by shaking level

VIII	63 thousand
VII	233 thousand
VI	3.7 million
V	5.3 million
IV	4.0 million

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



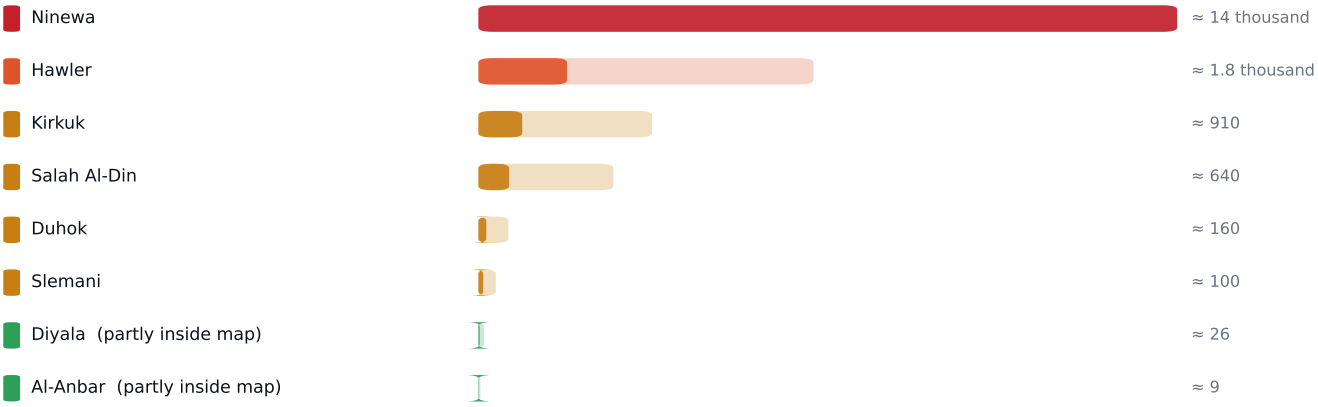
Little or no damage: 98% | Moderate damage: 2% | Heavy damage: <1% | Very heavy or collapse: <1%

Which buildings failed

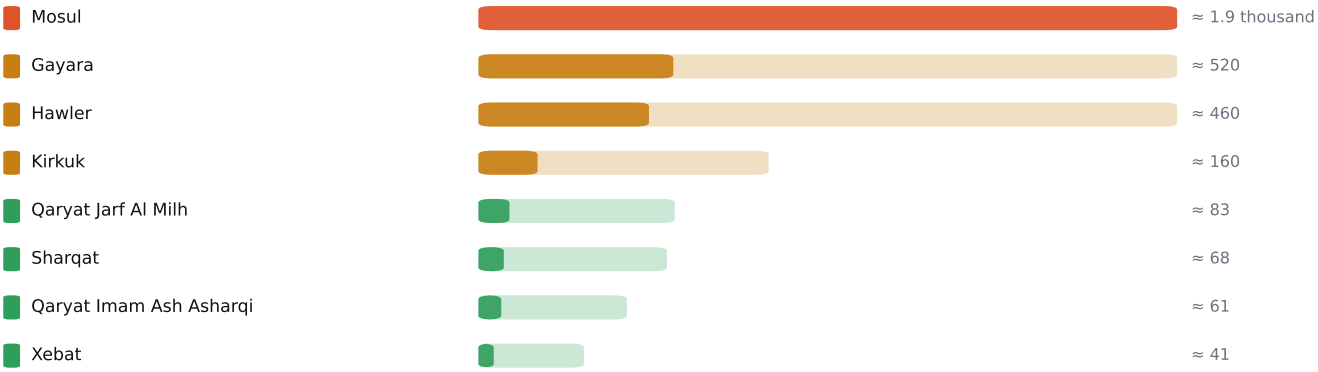
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**2.4%**	≈ 2.8 thousand buildings
Adobe, earth brick	class A	**1.8%**	≈ 2.2 thousand buildings
Simple stone	class B	**0.7%**	≈ 2.4 thousand buildings
Manufactured stone units, timber floors	class B	**0.6%**	≈ 3.3 thousand buildings
RC flat slab and others	class B	**0.5%**	≈ 16 buildings
RC frame — no / low code (ERD-L)	class C	**0.2%**	≈ 290 buildings
Massive stone	class C	**0.2%**	≈ 50 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Ninewa	1.0 million	11 thousand	4.6 thousand to 52 thousand	1.0%
Hawler	660 thousand	1.1 thousand	500 to 6.9 thousand	0.17%
Kirkuk	569 thousand	550	290 to 3.6 thousand	0.10%
Salah Al-Din	378 thousand	410	160 to 2.8 thousand	0.11%
Duhok	266 thousand	110	49 to 620	0.04%
Slemani	490 thousand	77	37 to 360	0.02%
Diyala	132 thousand	19	8 to 120	0.01%
Al-Anbar	65 thousand	7	3 to 41	0.01%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 0 felt-report cells (felt channel: none)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml19440001, version 2, generated 19 Oct 2026, 11:22 UTC.